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南 华 中 学

NAN HUA HIGH SCHOOL

2024 WEIGHTED ASSESSMENT 1

Subject : Mathematics
Paper : 4052
Level : Secondary Two Express
Date : 21 February 2024
Duration : 45 minutes

READ THESE INSTRUCTIONS FIRST

Write your name, index number and class on all the work you hand in.
 Write in dark blue or black pen.
 You may use an HB pencil for any diagrams or graphs.
 Do not use staples, paper clips, glue or correction fluid / tape.

Answer **all** the questions. The number of marks is given in brackets [] at the end of each question or part question.

If working is needed for any question, it must be shown with the answer.

Omission of essential working will result in loss of marks.

The total of the marks for this paper is 30.

The use of an approved scientific calculator is expected, where appropriate.

If the degree of accuracy is not specified in the question and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

For π , use either your calculator value or 3.142.

FOR MARKER'S USE ONLY		
Question Number	Maximum Mark	Mark
1	3	
2	4	
3	3	
4	7	
5	3	
6	4	
7	6	
Total Mark	30	

Answer **all** questions

- 1 Solve the following pair of simultaneous equations.

$$2x + 4y = 21$$

$$5x - 5y = 6$$

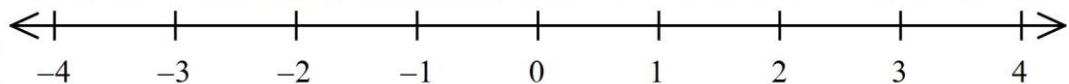
Answer $x = \dots\dots\dots$

$y = \dots\dots\dots$ [3]

- 2 (a) Solve the inequality $\frac{2(4m-6)}{-5} < 8$.

Answer [2]

- (b) Represent your solution in part (a) on the number line below. [1]



- (c) Using your solution in part (a), find the smallest possible value of m if m is an integer.

Answer [1]

3 Solve the following pair of simultaneous equations.

$$\frac{x-1}{y+1} = \frac{1}{5}$$

$$\frac{x+3}{y+3} = \frac{1}{3}$$

Answer $x = \dots\dots\dots$

$y = \dots\dots\dots$ [3]

- 4 (a) Expand and simplify $48ba - (7a - 3b)(2a + 9b)$

Answer [2]

- (b) Factorise each of the following expressions completely.

(i) $3p^2 - 14p + 15$

Answer [2]

(ii) $2x^3 - 10x^2y - 28xy^2$

Answer [3]

- 5 Mr Chiam bought some apples at \$2 each and some watermelons at \$5 each. He bought a total of 50 fruits, and the cost of the apples was at least \$20 more than the cost of the watermelons.
By **forming an inequality**, find the minimum number of apples that Mr Chiam bought.

Answer [3]

- 6 Line 1 and the graph of $2x - y = 2$ are drawn on the grid shown on the next page.

- (a) State the equation of Line 1.

Answer [1]

- (b) On the same axes, draw the graph of $2x + y = 4$. [2]

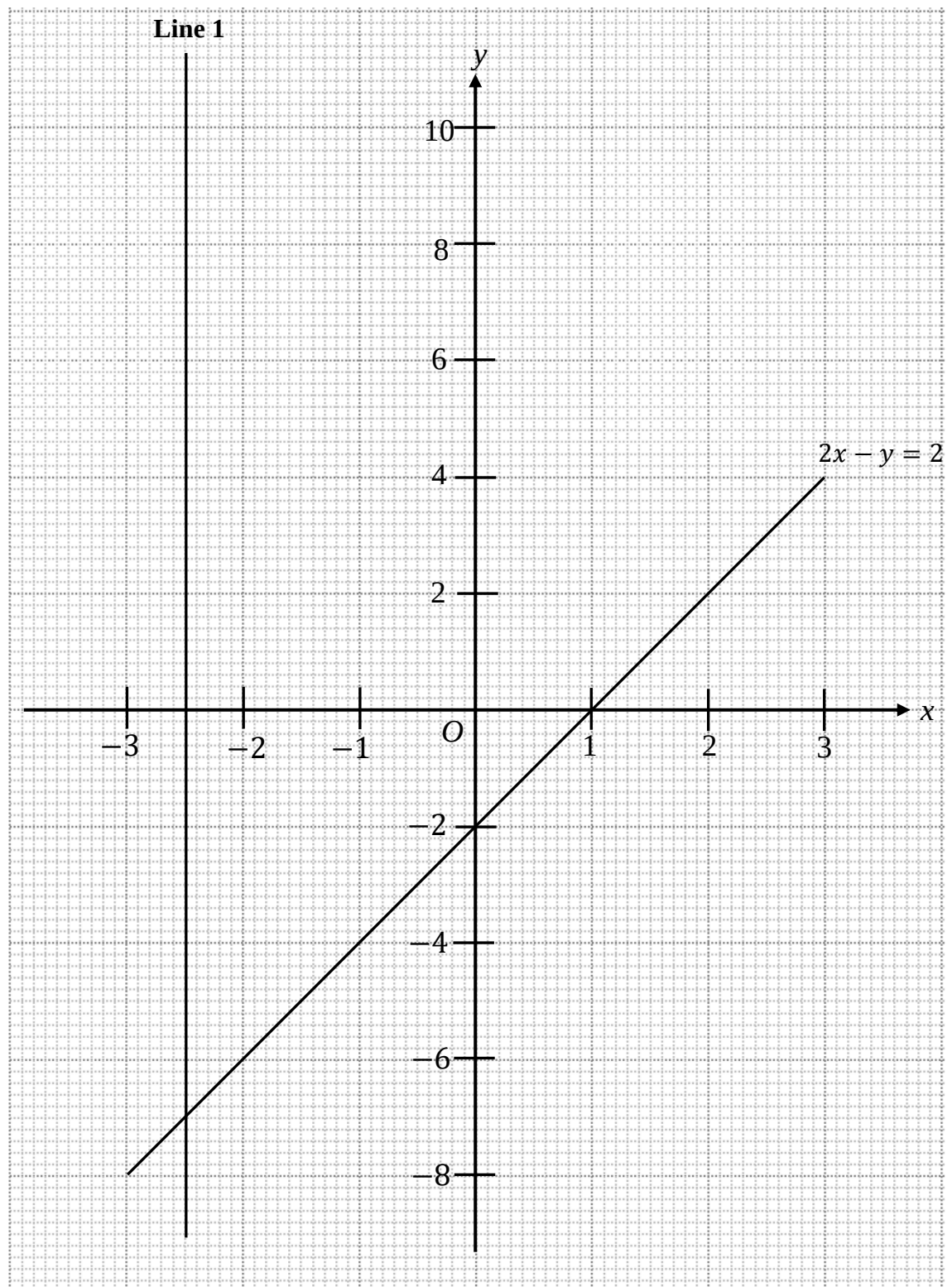
- (c) Hence, by using the **graphical method**, solve the simultaneous equations

$$2x - y = 2,$$

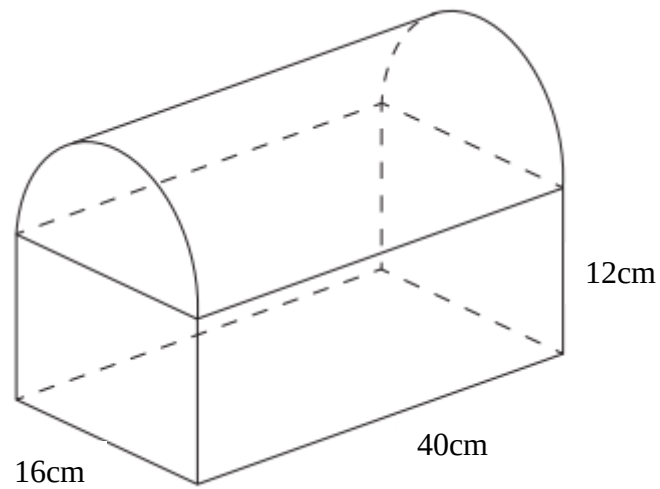
$$2x + y = 4.$$

Answer $x =$

$y =$ [1]



- 7 The figure below shows a solid that is made up of a half-cylinder and a cuboid with dimensions 16 cm by 40 cm by 12 cm.



- (a) Find the volume of the solid.

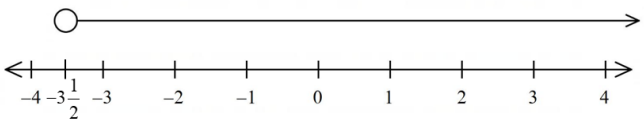
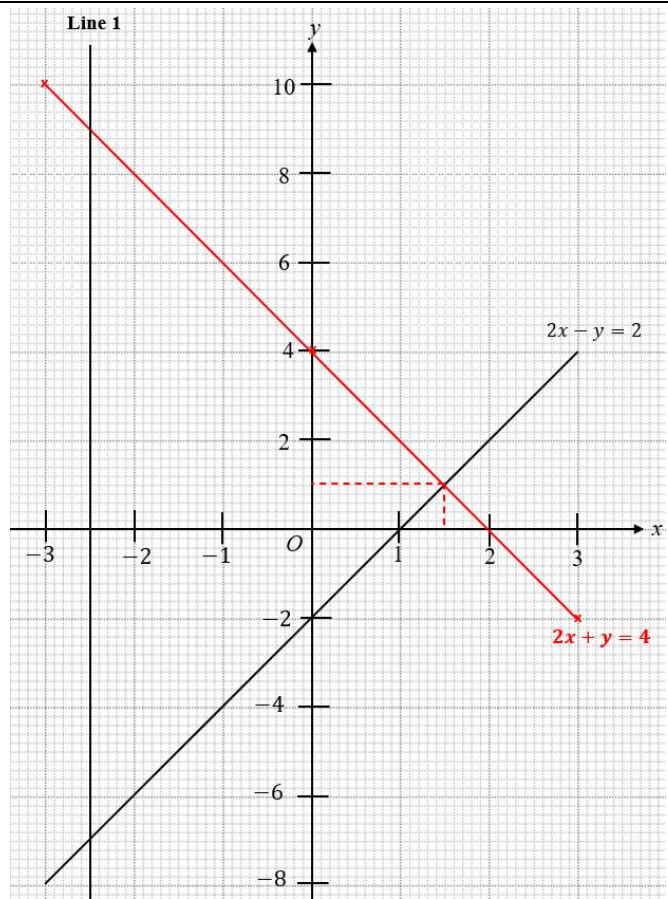
Answer cm^3 [3]

- (b) Find the total surface area of the solid.

Answer cm^2 [3]

End of Paper

Answer Key

1	$x = 4.3$ $y = 3.1$		
2a	$m > -3\frac{1}{2}$	2b	
2c	-3		
3a	$x = 6$ $y = 24$		
4a	$-14a^2 - 9ab + 27b^2$	4b(i)	$(3p - 5)(p - 3)$
		4b(ii)	$2x(x + 2y)(x - 7y)$
5	39		
6a	$x = -2.5$	6b	
6c	$x = 1.5$ $y = 1$		
7a	11700 cm^3	7b	3190 cm^2